

Milestone-1 Technical Documentation

Project: *Unified Military Analytics and Comparison Dashboard (DV)*

Phase: Data Collection & Preparation

1. Introduction

The goal of Milestone-1 is to build a **robust data collection and preparation pipeline** that gathers large-scale military and economic data for more than **140 countries** from GlobalFirepower.com.

This data serves as the foundation for all analytics, dashboards, KPIs, and coalition simulations that will be implemented in later milestones.

The entire pipeline is developed in **Python** to ensure automation, accuracy, and reproducibility.

2. Data Source

The project uses **GlobalFirepower.com**, which publishes verified military and economic statistics of all major nations.

A pre-defined file `links_for_military_data.txt` contains URLs for **50+ military indicators**, including:

- Personnel
- Aircraft
- Tanks
- Navy
- Defense budget
- Energy resources
- Infrastructure
- Geography

Each URL represents **one metric across all countries**.

3. Technology Stack

Component	Purpose
Python	Main programming language
Requests	Sends HTTP requests to web pages
BeautifulSoup	Parses and extracts HTML data
Regex (re)	Cleans numeric values
Pandas	Data transformation and CSV generation

4. Architecture of the Scraping System

The system follows a **URL-driven architecture**:

```
URL → Download HTML → Parse Country + Value → Store → Merge → Pivot → CSV
```

Each URL is mapped to a metric name using a Python dictionary:

```
Dict = {url : metric_name}
```

This ensures that all scraped values are automatically labeled with the correct column name.

5. Web Scraping Logic

5.1 Sending Requests

The `requests.get()` function is used to download HTML from each URL:

```
response = requests.get(u)
```

This retrieves the complete web page containing military statistics.

5.2 HTML Parsing

The HTML is converted into a navigable tree using BeautifulSoup:

```
soup = BeautifulSoup(response.text, 'html.parser')
```

5.3 Locating Country Data

All country records are embedded inside `<a>` tags containing `"country_id"`.

```
all_links = soup.find_all('a')
```

Only those links containing "country_id" are extracted.

5.4 Extracting Values

Each country block contains two key elements:

HTML Class	Meaning
longFormName	Country Name
valueContainer	Military Metric

Extraction:

```
country = country_div.text.strip()
value = value_div.text.strip()
```

6. Data Cleaning Process

Military values often contain:

- commas
- plus signs
- percentages
- spaces

Example: "1,245,000km"

These are removed using **Regular Expressions**:

```
value = re.sub("[^0-9]", "", value)
```

This ensures that **all data is numeric** and ready for mathematical operations.

7. Data Storage Structure

Each scraped entry is stored as:

```
{
  "country": "India",
  "metric": "active_personnel",
  "value": 1450000
}
```

This format allows merging all URLs into one master dataset.

8. Data Restructuring Using Pandas

After scraping, the dataset is reshaped into an **analytics-ready table** using:

```
df = df.pivot(index="country", columns="metric", values="value")
```

This converts the dataset into:

country tanks fighters budget manpower ...

Each country becomes one row, and each metric becomes one column.

9. Handling Missing Data

Some metrics are unavailable for certain countries.

These appear as `NaN` values in Pandas.

Non-useful columns like naval tonnage were removed:

```
df.drop('total_naval_fleet_tonnage_mt', axis=1)
```

This keeps the dataset clean and consistent.

10. Final Outputs

Two datasets are generated:

File	Description
<code>military_raw_data.csv</code>	Raw scraped data
<code>military_cleaned.csv</code>	Cleaned, structured analytics data

These files are ready for:

- Tableau
- Power BI
- Dash